SPARKY'S SCIENCE EXPERIMENTS

Box Solar Oven: Cook with Sunlight

Physical Science (energy) · Ages 8+ · 1 hour, on a sunny day

Build an oven from a cardboard box, foil and plastic that uses only sunshine to melt chocolate and warm food.

Safety first

Sparky always says: The oven gets hot inside - always use an oven glove to take food out!

- An adult must cut the box. Don't stare at the shiny foil in the sun.
- Only warm food that is safe to eat cold, like bread or chocolate - never raw meat or eggs.

You need

- A box with a flap lid (pizza box or shoe box)
- Aluminium foil
- Cling film (clear plastic wrap)
- Black paper or a black plastic bag
- Tape, a ruler, a stick, and a craft knife (adult)

Steps

1. Draw a square on the lid, 3 cm in from the edges. An adult cuts three sides to make a flap.
2. Cover the inside of the flap with foil, shiny side out. Tape it flat.
3. Tape cling film tightly over the hole under the flap, as a window.
4. Line the bottom of the box with black paper and put the food on it.
5. Close the lid, face the box to the sun (north around midday in South Africa) and prop the flap with the stick so it shines light through the window.
6. Wait 30 to 60 minutes, turning the box to follow the sun.

If it doesn't work, try these in order

1. Not getting hot? It needs a hot day with direct sun, best between 11:00 and 15:00.
2. Check the cling film is tight with no gaps, so warm air can't escape.
3. Turn the box to face the sun every 20 minutes.

Why it works

The foil reflects extra sunlight into the box. The black paper absorbs the light and turns it into heat. The plastic window lets sunlight in but traps the warm air, like a car in the sun. A box oven can reach about 65 to 90 degrees Celsius on a hot day.

Did you know? South Africa gets about 2,500 hours of sunshine a year on average - that is why many big solar power stations have been built in the sunny Northern Cape.

Ask Sparky: Why is the inside black and the flap shiny?